Frozen Warning Decision Comparisons Complete Supplement

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This supplement retains every specified frozen-policy comparison: 720 policy-minus-no-alarm settings and 540 XGBoost-minus-grade settings. Record blocks 1-4 precede supplementary pooled sums. Stage 1 and Stage 2 denote research stages; blocks denote evaluated record groups.

Negative loss differences favor the left policy. Costs are assumed relative values, not measured money or accident avoidance. All intervals are unadjusted conditional 95% paired moving-block intervals from 2,000 draws of length 32 within blocks; they exclude model fitting, tuning and rule selection. Identical decisions can yield exact zero intervals. No policy is reselected from these tables.

Integrated manuscript in this edition

Comparator protocol and fixed selection rules

Complete machine-readable evidence and verification

Policies minus no alarms

Table S1. Policy minus no alarms; assumed missed-hazard cost r=5, false-alert cost 1.

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
1	XGBoost	A_b01	1.55 [0.39, 3.10]	0	36	8	8	3
1	XGBoost	A_b05	1.94 [0.58, 3.88]	0	36	10	10	0
1	XGBoost	A_b10	2.13 [0.78, 4.07]	0	36	11	11	0
1	XGBoost	A_b20	1.74 [-0.58, 4.26]	2	34	19	21	0
1	XGBoost	B_r05	3.29 [-2.91, 8.91]	8	28	57	65	-
1	XGBoost	C_b01_r05	0.00 [0.00, 0.00]	0	36	0	0	0
1	XGBoost	C_b05_r05	1.55 [0.39, 3.10]	0	36	8	8	0
1	XGBoost	C_b10_r05	1.55 [0.39, 3.10]	0	36	8	8	0
1	XGBoost	C_b20_r05	2.13 [-0.78, 4.85]	1	35	16	17	0
1	ghazard	A_b01	0.39 [-6.20, 5.62]	4	32	22	26	21
1	ghazard	A_b05	0.39 [-6.20, 5.62]	4	32	22	26	1
1	ghazard	A_b10	0.39 [-6.20, 5.62]	4	32	22	26	0
1	ghazard	A_b20	0.39 [-6.20, 5.62]	4	32	22	26	0
1	ghazard	B_r05	58.14 [31.40, 81.40]	36	0	480	516	-
1	ghazard	C_b01_r05	0.39 [-6.20, 5.62]	4	32	22	26	21

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
1	ghazard	C_b05_r05	0.39 [-6.20, 5.62]	4	32	22	26	1
1	ghazard	C_b10_r05	0.39 [-6.20, 5.62]	4	32	22	26	0
1	ghazard	C_b20_r05	0.39 [-6.20, 5.62]	4	32	22	26	0
1	seismic	A_b01	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	B_r05	-0.78 [-12.40, 7.56]	15	21	71	86	-
1	seismic	C_b01_r05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b05_r05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b10_r05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b20_r05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	B_r05	20.93 [9.69, 34.88]	11	25	163	174	-
1	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b05_r05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b10_r05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b20_r05	0.00 [0.00, 0.00]	0	36	0	0	0
2	XGBoost	A_b01	7.18 [3.50, 9.51]	0	10	37	37	32
2	XGBoost	A_b05	13.01 [7.18, 17.28]	0	10	67	67	42
2	XGBoost	A_b10	18.45 [10.29, 25.24]	3	7	110	113	62
2	XGBoost	A_b20	42.33 [33.20, 53.79]	6	4	248	254	151
2	XGBoost	B_r05	0.00 [0.00, 0.00]	0	10	0	0	-
2	XGBoost	C_b01_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b05_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b10_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b20_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b05	0.00 [0.00, 0.00]	0	10	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
2	ghazard	A_b10	10.10 [-1.36, 21.17]	2	8	62	64	13
2	ghazard	A_b20	10.10 [-1.36, 21.17]	2	8	62	64	0
2	ghazard	B_r05	0.00 [0.00, 0.00]	0	10	0	0	-
2	ghazard	C_b01_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b05_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b10_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b20_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	A_b05	18.25 [5.83, 30.49]	6	4	124	130	105
2	seismic	A_b10	18.25 [5.83, 30.49]	6	4	124	130	79
2	seismic	A_b20	18.25 [5.83, 30.49]	6	4	124	130	27
2	seismic	B_r05	0.00 [0.00, 0.00]	0	10	0	0	-
2	seismic	C_b01_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b05_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b10_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b20_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	B_r05	0.00 [0.00, 0.00]	0	10	0	0	-
2	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b05_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b10_r05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b20_r05	0.00 [0.00, 0.00]	0	10	0	0	0
3	XGBoost	A_b01	1.55 [0.19, 3.68]	0	24	8	8	3
3	XGBoost	A_b05	3.68 [-1.36, 6.98]	6	18	49	55	30
3	XGBoost	A_b10	5.62 [0.39, 8.91]	7	17	64	71	20
3	XGBoost	A_b20	21.32 [14.34, 27.33]	10	14	160	170	67
3	XGBoost	B_r05	0.00 [0.00, 0.00]	0	24	0	0	-
3	XGBoost	C_b01_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b05_r05	0.00 [0.00, 0.00]	0	24	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
3	XGBoost	C_b10_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b20_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b20	12.79 [4.65, 20.93]	3	21	81	84	0
3	ghazard	B_r05	0.00 [0.00, 0.00]	0	24	0	0	-
3	ghazard	C_b01_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b05_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b10_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b20_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	B_r05	0.00 [0.00, 0.00]	0	24	0	0	-
3	seismic	C_b01_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b05_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b10_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b20_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	B_r05	0.00 [0.00, 0.00]	0	24	0	0	-
3	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b05_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b10_r05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b20_r05	0.00 [0.00, 0.00]	0	24	0	0	0
4	XGBoost	A_b01	1.16 [0.39, 2.33]	0	18	6	6	1
4	XGBoost	A_b05	3.10 [-0.78, 6.98]	3	15	31	34	9
4	XGBoost	A_b10	5.62 [1.36, 10.86]	4	14	49	53	2

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
4	XGBoost	A_b20	13.57 [6.01, 20.93]	7	11	105	112	9
4	XGBoost	B_r05	2.13 [0.00, 4.46]	1	17	16	17	-
4	XGBoost	C_b01_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	XGBoost	C_b05_r05	2.13 [0.00, 4.46]	1	17	16	17	0
4	XGBoost	C_b10_r05	2.13 [0.00, 4.46]	1	17	16	17	0
4	XGBoost	C_b20_r05	2.13 [0.00, 4.46]	1	17	16	17	0
4	ghazard	A_b01	5.43 [0.00, 15.70]	0	18	28	28	23
4	ghazard	A_b05	5.43 [0.00, 15.70]	0	18	28	28	3
4	ghazard	A_b10	5.43 [0.00, 15.70]	0	18	28	28	0
4	ghazard	A_b20	8.72 [1.74, 19.77]	2	16	55	57	0
4	ghazard	B_r05	0.00 [0.00, 0.00]	0	18	0	0	-
4	ghazard	C_b01_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b05_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b10_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b20_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b01	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	B_r05	0.00 [0.00, 0.00]	0	18	0	0	-
4	seismic	C_b01_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b05_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b10_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b20_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	B_r05	0.00 [0.00, 0.00]	0	18	0	0	-
4	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b05_r05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b10_r05	0.00 [0.00, 0.00]	0	18	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
4	seismoacoustic	C_b20_r05	0.00 [0.00, 0.00]	0	18	0	0	0
Pooled	XGBoost	A_b01	2.86 [1.75, 3.69]	0	88	59	59	39
Pooled	XGBoost	A_b05	5.43 [2.96, 6.93]	9	79	157	166	81
Pooled	XGBoost	A_b10	7.95 [4.94, 10.13]	14	74	234	248	84
Pooled	XGBoost	A_b20	19.73 [16.19, 23.56]	25	63	532	557	227
Pooled	XGBoost	B_r05	1.36 [-0.24, 2.96]	9	79	73	82	-
Pooled	XGBoost	C_b01_r05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	XGBoost	C_b05_r05	0.92 [0.34, 1.65]	1	87	24	25	0
Pooled	XGBoost	C_b10_r05	0.92 [0.34, 1.65]	1	87	24	25	0
Pooled	XGBoost	C_b20_r05	1.07 [0.15, 1.99]	2	86	32	34	0
Pooled	ghazard	A_b01	1.45 [-0.73, 4.22]	4	84	50	54	44
Pooled	ghazard	A_b05	1.45 [-0.73, 4.22]	4	84	50	54	4
Pooled	ghazard	A_b10	3.97 [0.29, 7.85]	6	82	112	118	13
Pooled	ghazard	A_b20	8.00 [3.49, 12.22]	11	77	220	231	0
Pooled	ghazard	B_r05	14.54 [7.56, 20.36]	36	52	480	516	-
Pooled	ghazard	C_b01_r05	0.10 [-1.55, 1.41]	4	84	22	26	21
Pooled	ghazard	C_b05_r05	0.10 [-1.55, 1.41]	4	84	22	26	1
Pooled	ghazard	C_b10_r05	0.10 [-1.55, 1.41]	4	84	22	26	0
Pooled	ghazard	C_b20_r05	0.10 [-1.55, 1.41]	4	84	22	26	0
Pooled	seismic	A_b01	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	A_b05	4.56 [1.50, 7.71]	6	82	124	130	105
Pooled	seismic	A_b10	4.56 [1.50, 7.71]	6	82	124	130	79
Pooled	seismic	A_b20	4.56 [1.50, 7.71]	6	82	124	130	27
Pooled	seismic	B_r05	-0.19 [-3.20, 1.94]	15	73	71	86	-
Pooled	seismic	C_b01_r05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b05_r05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b10_r05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b20_r05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	88	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
Pooled	seismoacoustic	B_r05	5.24 [2.28, 8.77]	11	77	163	174	-
Pooled	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b05_r05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b10_r05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b20_r05	0.00 [0.00, 0.00]	0	88	0	0	0

Table S2. Policy minus no alarms; assumed missed-hazard cost r=10, false-alert cost 1.

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
1	XGBoost	A_b01	1.55 [0.39, 3.10]	0	36	8	8	3
1	XGBoost	A_b05	1.94 [0.58, 3.88]	0	36	10	10	0
1	XGBoost	A_b10	2.13 [0.78, 4.07]	0	36	11	11	0
1	XGBoost	A_b20	-0.19 [-5.23, 3.69]	2	34	19	21	0
1	XGBoost	B_r10	4.46 [-39.73, 41.48]	34	2	363	397	-
1	XGBoost	C_b01_r10	0.00 [0.00, 0.00]	0	36	0	0	0
1	XGBoost	C_b05_r10	1.55 [0.39, 3.10]	0	36	8	8	0
1	XGBoost	C_b10_r10	2.13 [0.78, 4.07]	0	36	11	11	0
1	XGBoost	C_b20_r10	1.16 [-3.88, 4.84]	1	35	16	17	0
1	ghazard	A_b01	-3.49 [-16.09, 5.43]	4	32	22	26	21
1	ghazard	A_b05	-3.49 [-16.09, 5.43]	4	32	22	26	1
1	ghazard	A_b10	-3.49 [-16.09, 5.43]	4	32	22	26	0
1	ghazard	A_b20	-3.49 [-16.09, 5.43]	4	32	22	26	0
1	ghazard	B_r10	23.26 [-25.78, 65.89]	36	0	480	516	-
1	ghazard	C_b01_r10	-3.49 [-16.09, 5.43]	4	32	22	26	21
1	ghazard	C_b05_r10	-3.49 [-16.09, 5.43]	4	32	22	26	1
1	ghazard	C_b10_r10	-3.49 [-16.09, 5.43]	4	32	22	26	0
1	ghazard	C_b20_r10	-3.49 [-16.09, 5.43]	4	32	22	26	0
1	seismic	A_b01	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	B_r10	23.26 [-25.78, 65.89]	36	0	480	516	-
1	seismic	C_b01_r10	0.00 [0.00, 0.00]	0	36	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
1	seismic	C_b05_r10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b10_r10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b20_r10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	B_r10	23.26 [-25.78, 65.89]	36	0	480	516	-
1	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b05_r10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b10_r10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b20_r10	0.00 [0.00, 0.00]	0	36	0	0	0
2	XGBoost	A_b01	7.18 [3.50, 9.51]	0	10	37	37	32
2	XGBoost	A_b05	13.01 [7.18, 17.28]	0	10	67	67	42
2	XGBoost	A_b10	15.53 [4.47, 23.69]	3	7	110	113	62
2	XGBoost	A_b20	36.50 [27.18, 49.13]	6	4	248	254	151
2	XGBoost	B_r10	0.00 [0.00, 0.00]	0	10	0	0	-
2	XGBoost	C_b01_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b05_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b10_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b20_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b05	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b10	8.16 [-5.64, 20.00]	2	8	62	64	13
2	ghazard	A_b20	8.16 [-5.64, 20.00]	2	8	62	64	0
2	ghazard	B_r10	0.00 [0.00, 0.00]	0	10	0	0	-
2	ghazard	C_b01_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b05_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b10_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b20_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	A_b05	12.43 [1.94, 23.89]	6	4	124	130	105

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
2	seismic	A_b10	12.43 [1.94, 23.89]	6	4	124	130	79
2	seismic	A_b20	12.43 [1.94, 23.89]	6	4	124	130	27
2	seismic	B_r10	0.00 [0.00, 0.00]	0	10	0	0	-
2	seismic	C_b01_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b05_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b10_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b20_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	B_r10	0.00 [0.00, 0.00]	0	10	0	0	-
2	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b05_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b10_r10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b20_r10	0.00 [0.00, 0.00]	0	10	0	0	0
3	XGBoost	A_b01	1.55 [0.19, 3.68]	0	24	8	8	3
3	XGBoost	A_b05	-2.13 [-10.66, 3.49]	6	18	49	55	30
3	XGBoost	A_b10	-1.16 [-10.86, 5.04]	7	17	64	71	20
3	XGBoost	A_b20	11.63 [0.00, 21.90]	10	14	160	170	67
3	XGBoost	B_r10	0.00 [0.00, 0.00]	0	24	0	0	-
3	XGBoost	C_b01_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b05_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b10_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b20_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b20	9.88 [3.49, 16.67]	3	21	81	84	0
3	ghazard	B_r10	0.00 [0.00, 0.00]	0	24	0	0	-
3	ghazard	C_b01_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b05_r10	0.00 [0.00, 0.00]	0	24	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
3	ghazard	C_b10_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b20_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	B_r10	0.00 [0.00, 0.00]	0	24	0	0	-
3	seismic	C_b01_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b05_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b10_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b20_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	B_r10	0.00 [0.00, 0.00]	0	24	0	0	-
3	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b05_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b10_r10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b20_r10	0.00 [0.00, 0.00]	0	24	0	0	0
4	XGBoost	A_b01	1.16 [0.39, 2.33]	0	18	6	6	1
4	XGBoost	A_b05	0.19 [-7.76, 6.40]	3	15	31	34	9
4	XGBoost	A_b10	1.74 [-6.40, 9.30]	4	14	49	53	2
4	XGBoost	A_b20	6.78 [-5.62, 16.67]	7	11	105	112	9
4	XGBoost	B_r10	1.16 [-3.10, 4.46]	1	17	16	17	-
4	XGBoost	C_b01_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	XGBoost	C_b05_r10	1.16 [-3.10, 4.46]	1	17	16	17	0
4	XGBoost	C_b10_r10	1.16 [-3.10, 4.46]	1	17	16	17	0
4	XGBoost	C_b20_r10	1.16 [-3.10, 4.46]	1	17	16	17	0
4	ghazard	A_b01	5.43 [0.00, 15.70]	0	18	28	28	23
4	ghazard	A_b05	5.43 [0.00, 15.70]	0	18	28	28	3
4	ghazard	A_b10	5.43 [0.00, 15.70]	0	18	28	28	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
4	ghazard	A_b20	6.78 [0.78, 16.47]	2	16	55	57	0
4	ghazard	B_r10	0.00 [0.00, 0.00]	0	18	0	0	-
4	ghazard	C_b01_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b05_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b10_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b20_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b01	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	B_r10	0.00 [0.00, 0.00]	0	18	0	0	-
4	seismic	C_b01_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b05_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b10_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b20_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	B_r10	0.00 [0.00, 0.00]	0	18	0	0	-
4	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b05_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b10_r10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b20_r10	0.00 [0.00, 0.00]	0	18	0	0	0
Pooled	XGBoost	A_b01	2.86 [1.75, 3.69]	0	88	59	59	39
Pooled	XGBoost	A_b05	3.25 [-0.29, 5.38]	9	79	157	166	81
Pooled	XGBoost	A_b10	4.56 [-0.00, 7.51]	14	74	234	248	84
Pooled	XGBoost	A_b20	13.67 [8.77, 18.61]	25	63	532	557	227
Pooled	XGBoost	B_r10	1.41 [-9.26, 11.05]	35	53	379	414	-
Pooled	XGBoost	C_b01_r10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	XGBoost	C_b05_r10	0.68 [-0.34, 1.65]	1	87	24	25	0
Pooled	XGBoost	C_b10_r10	0.82 [-0.19, 1.84]	1	87	27	28	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
Pooled	XGBoost	C_b20_r10	0.58 [-1.02, 1.89]	2	86	32	34	0
Pooled	ghazard	A_b01	0.48 [-3.05, 3.88]	4	84	50	54	44
Pooled	ghazard	A_b05	0.48 [-3.05, 3.88]	4	84	50	54	4
Pooled	ghazard	A_b10	2.52 [-2.38, 7.13]	6	82	112	118	13
Pooled	ghazard	A_b20	5.33 [-0.20, 9.70]	11	77	220	231	0
Pooled	ghazard	B_r10	5.82 [-6.98, 16.48]	36	52	480	516	-
Pooled	ghazard	C_b01_r10	-0.87 [-3.97, 1.36]	4	84	22	26	21
Pooled	ghazard	C_b05_r10	-0.87 [-3.97, 1.36]	4	84	22	26	1
Pooled	ghazard	C_b10_r10	-0.87 [-3.97, 1.36]	4	84	22	26	0
Pooled	ghazard	C_b20_r10	-0.87 [-3.97, 1.36]	4	84	22	26	0
Pooled	seismic	A_b01	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	A_b05	3.10 [0.48, 6.06]	6	82	124	130	105
Pooled	seismic	A_b10	3.10 [0.48, 6.06]	6	82	124	130	79
Pooled	seismic	A_b20	3.10 [0.48, 6.06]	6	82	124	130	27
Pooled	seismic	B_r10	5.82 [-6.98, 16.48]	36	52	480	516	-
Pooled	seismic	C_b01_r10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b05_r10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b10_r10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b20_r10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	B_r10	5.82 [-6.98, 16.48]	36	52	480	516	-
Pooled	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b05_r10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b10_r10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b20_r10	0.00 [0.00, 0.00]	0	88	0	0	0

Table S3. Policy minus no alarms; assumed missed-hazard cost r=20, false-alert cost 1.

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
1	XGBoost	A_b01	1.55 [0.39, 3.10]	0	36	8	8	3
1	XGBoost	A_b05	1.94 [0.58, 3.88]	0	36	10	10	0
1	XGBoost	A_b10	2.13 [0.78, 4.07]	0	36	11	11	0
1	XGBoost	A_b20	-4.07 [-15.12, 3.68]	2	34	19	21	0
1	XGBoost	B_r20	-61.43 [-149.25, 11.64]	34	2	363	397	-
1	XGBoost	C_b01_r20	0.00 [0.00, 0.00]	0	36	0	0	0
1	XGBoost	C_b05_r20	1.55 [0.39, 3.10]	0	36	8	8	0
1	XGBoost	C_b10_r20	2.13 [0.78, 4.07]	0	36	11	11	0
1	XGBoost	C_b20_r20	-0.78 [-10.08, 4.84]	1	35	16	17	0
1	ghazard	A_b01	-11.24 [-35.47, 5.43]	4	32	22	26	21
1	ghazard	A_b05	-11.24 [-35.47, 5.43]	4	32	22	26	1
1	ghazard	A_b10	-11.24 [-35.47, 5.43]	4	32	22	26	0
1	ghazard	A_b20	-11.24 [-35.47, 5.43]	4	32	22	26	0
1	ghazard	B_r20	-46.51 [-140.12, 34.88]	36	0	480	516	-
1	ghazard	C_b01_r20	-11.24 [-35.47, 5.43]	4	32	22	26	21
1	ghazard	C_b05_r20	-11.24 [-35.47, 5.43]	4	32	22	26	1
1	ghazard	C_b10_r20	-11.24 [-35.47, 5.43]	4	32	22	26	0
1	ghazard	C_b20_r20	-11.24 [-35.47, 5.43]	4	32	22	26	0
1	seismic	A_b01	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	B_r20	-46.51 [-140.12, 34.88]	36	0	480	516	-
1	seismic	C_b01_r20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b05_r20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b10_r20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b20_r20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	36	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
1	seismoacoustic	B_r20	-46.51 [-140.12, 34.88]	36	0	480	516	-
1	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b05_r20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b10_r20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b20_r20	0.00 [0.00, 0.00]	0	36	0	0	0
2	XGBoost	A_b01	7.18 [3.50, 9.51]	0	10	37	37	32
2	XGBoost	A_b05	13.01 [7.18, 17.28]	0	10	67	67	42
2	XGBoost	A_b10	9.71 [-9.32, 22.52]	3	7	110	113	62
2	XGBoost	A_b20	24.85 [9.13, 43.50]	6	4	248	254	151
2	XGBoost	B_r20	0.00 [0.00, 0.00]	0	10	0	0	-
2	XGBoost	C_b01_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b05_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b10_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b20_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b05	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b10	4.27 [-16.89, 19.03]	2	8	62	64	13
2	ghazard	A_b20	4.27 [-16.89, 19.03]	2	8	62	64	0
2	ghazard	B_r20	0.00 [0.00, 0.00]	0	10	0	0	-
2	ghazard	C_b01_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b05_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b10_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b20_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	A_b05	0.78 [-13.99, 14.76]	6	4	124	130	105
2	seismic	A_b10	0.78 [-13.99, 14.76]	6	4	124	130	79
2	seismic	A_b20	0.78 [-13.99, 14.76]	6	4	124	130	27
2	seismic	B_r20	0.00 [0.00, 0.00]	0	10	0	0	-
2	seismic	C_b01_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b05_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b10_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b20_r20	0.00 [0.00, 0.00]	0	10	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
2	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	B_r20	0.00 [0.00, 0.00]	0	10	0	0	-
2	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b05_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b10_r20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b20_r20	0.00 [0.00, 0.00]	0	10	0	0	0
3	XGBoost	A_b01	1.55 [0.19, 3.68]	0	24	8	8	3
3	XGBoost	A_b05	-13.76 [-30.43, -2.13]	6	18	49	55	30
3	XGBoost	A_b10	-14.73 [-34.30, -1.35]	7	17	64	71	20
3	XGBoost	A_b20	-7.75 [-29.46, 12.21]	10	14	160	170	67
3	XGBoost	B_r20	0.00 [0.00, 0.00]	0	24	0	0	-
3	XGBoost	C_b01_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b05_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b10_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b20_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b20	4.07 [-2.91, 11.05]	3	21	81	84	0
3	ghazard	B_r20	0.00 [0.00, 0.00]	0	24	0	0	-
3	ghazard	C_b01_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b05_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b10_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b20_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	B_r20	3.10 [-24.23, 26.94]	12	12	256	268	-

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
3	seismic	C_b01_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b05_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b10_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b20_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	B_r20	0.00 [0.00, 0.00]	0	24	0	0	-
3	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b05_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b10_r20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b20_r20	0.00 [0.00, 0.00]	0	24	0	0	0
4	XGBoost	A_b01	1.16 [0.39, 2.33]	0	18	6	6	1
4	XGBoost	A_b05	-5.62 [-22.67, 5.81]	3	15	31	34	9
4	XGBoost	A_b10	-6.01 [-22.68, 7.37]	4	14	49	53	2
4	XGBoost	A_b20	-6.78 [-30.43, 11.63]	7	11	105	112	9
4	XGBoost	B_r20	13.18 [-15.50, 34.31]	12	6	308	320	-
4	XGBoost	C_b01_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	XGBoost	C_b05_r20	-0.78 [-9.69, 4.46]	1	17	16	17	0
4	XGBoost	C_b10_r20	-6.01 [-22.68, 7.37]	4	14	49	53	2
4	XGBoost	C_b20_r20	-6.01 [-22.68, 7.37]	4	14	49	53	0
4	ghazard	A_b01	5.43 [0.00, 15.70]	0	18	28	28	23
4	ghazard	A_b05	5.43 [0.00, 15.70]	0	18	28	28	3
4	ghazard	A_b10	5.43 [0.00, 15.70]	0	18	28	28	0
4	ghazard	A_b20	2.91 [-6.98, 13.38]	2	16	55	57	0
4	ghazard	B_r20	26.74 [-9.88, 55.23]	18	0	498	516	-
4	ghazard	C_b01_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b05_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b10_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b20_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b01	0.00 [0.00, 0.00]	0	18	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
4	seismic	A_b05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	B_r20	26.74 [-9.88, 55.23]	18	0	498	516	-
4	seismic	C_b01_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b05_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b10_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b20_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	B_r20	2.52 [-16.09, 19.96]	7	11	153	160	-
4	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b05_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b10_r20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b20_r20	0.00 [0.00, 0.00]	0	18	0	0	0
Pooled	XGBoost	A_b01	2.86 [1.75, 3.69]	0	88	59	59	39
Pooled	XGBoost	A_b05	-1.11 [-7.46, 3.06]	9	79	157	166	81
Pooled	XGBoost	A_b10	-2.23 [-10.95, 3.30]	14	74	234	248	84
Pooled	XGBoost	A_b20	1.55 [-7.51, 10.23]	25	63	532	557	227
Pooled	XGBoost	B_r20	-12.07 [-35.68, 7.27]	46	42	671	717	-
Pooled	XGBoost	C_b01_r20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	XGBoost	C_b05_r20	0.19 [-1.79, 1.65]	1	87	24	25	0
Pooled	XGBoost	C_b10_r20	-0.97 [-5.14, 2.18]	4	84	60	64	2
Pooled	XGBoost	C_b20_r20	-1.70 [-6.69, 1.84]	5	83	65	70	0
Pooled	ghazard	A_b01	-1.45 [-7.90, 3.44]	4	84	50	54	44
Pooled	ghazard	A_b05	-1.45 [-7.90, 3.44]	4	84	50	54	4
Pooled	ghazard	A_b10	-0.39 [-8.48, 5.96]	6	82	112	118	13
Pooled	ghazard	A_b20	0.00 [-8.58, 6.59]	11	77	220	231	0
Pooled	ghazard	B_r20	-4.94 [-31.41, 16.43]	54	34	978	1032	-
Pooled	ghazard	C_b01_r20	-2.81 [-8.82, 1.36]	4	84	22	26	21

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
Pooled	ghazard	C_b05_r20	-2.81 [-8.82, 1.36]	4	84	22	26	1
Pooled	ghazard	C_b10_r20	-2.81 [-8.82, 1.36]	4	84	22	26	0
Pooled	ghazard	C_b20_r20	-2.81 [-8.82, 1.36]	4	84	22	26	0
Pooled	seismic	A_b01	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	A_b05	0.19 [-3.35, 3.78]	6	82	124	130	105
Pooled	seismic	A_b10	0.19 [-3.35, 3.78]	6	82	124	130	79
Pooled	seismic	A_b20	0.19 [-3.35, 3.78]	6	82	124	130	27
Pooled	seismic	B_r20	-4.17 [-31.90, 18.33]	66	22	1234	1300	-
Pooled	seismic	C_b01_r20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b05_r20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b10_r20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b20_r20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	B_r20	-11.00 [-35.63, 9.50]	43	45	633	676	-
Pooled	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b05_r20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b10_r20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b20_r20	0.00 [0.00, 0.00]	0	88	0	0	0

Table S4. Policy minus no alarms; assumed missed-hazard cost r=50, false-alert cost 1.

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
1	XGBoost	A_b01	1.55 [0.39, 3.10]	0	36	8	8	3
1	XGBoost	A_b05	1.94 [0.58, 3.88]	0	36	10	10	0
1	XGBoost	A_b10	2.13 [0.78, 4.07]	0	36	11	11	0
1	XGBoost	A_b20	-15.70 [-44.19, 3.68]	2	34	19	21	0
1	XGBoost	B_r50	-259.11 [-485.28, -75.51]	34	2	363	397	-
1	XGBoost	C_b01_r50	0.00 [0.00, 0.00]	0	36	0	0	0
1	XGBoost	C_b05_r50	1.55 [0.39, 3.10]	0	36	8	8	0
1	XGBoost	C_b10_r50	2.13 [0.78, 4.07]	0	36	11	11	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
1	XGBoost	C_b20_r50	-6.59 [-27.52, 4.84]	1	35	16	17	0
1	ghazard	A_b01	-34.50 [-93.60, 5.43]	4	32	22	26	21
1	ghazard	A_b05	-34.50 [-93.60, 5.43]	4	32	22	26	1
1	ghazard	A_b10	-34.50 [-93.60, 5.43]	4	32	22	26	0
1	ghazard	A_b20	-34.50 [-93.60, 5.43]	4	32	22	26	0
1	ghazard	B_r50	-255.81 [-483.14, -58.14]	36	0	480	516	-
1	ghazard	C_b01_r50	-34.50 [-93.60, 5.43]	4	32	22	26	21
1	ghazard	C_b05_r50	-34.50 [-93.60, 5.43]	4	32	22	26	1
1	ghazard	C_b10_r50	-34.50 [-93.60, 5.43]	4	32	22	26	0
1	ghazard	C_b20_r50	-34.50 [-93.60, 5.43]	4	32	22	26	0
1	seismic	A_b01	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	A_b20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	B_r50	-255.81 [-483.14, -58.14]	36	0	480	516	-
1	seismic	C_b01_r50	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b05_r50	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b10_r50	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismic	C_b20_r50	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	B_r50	-255.81 [-483.14, -58.14]	36	0	480	516	-
1	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b05_r50	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b10_r50	0.00 [0.00, 0.00]	0	36	0	0	0
1	seismoacoustic	C_b20_r50	0.00 [0.00, 0.00]	0	36	0	0	0
2	XGBoost	A_b01	7.18 [3.50, 9.51]	0	10	37	37	32
2	XGBoost	A_b05	13.01 [7.18, 17.28]	0	10	67	67	42
2	XGBoost	A_b10	-7.77 [-53.40, 21.75]	3	7	110	113	62
2	XGBoost	A_b20	-10.10 [-53.79, 34.56]	6	4	248	254	151

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
2	XGBoost	B_r50	0.00 [0.00, 0.00]	0	10	0	0	-
2	XGBoost	C_b01_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b05_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b10_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	XGBoost	C_b20_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b05	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	A_b10	-7.38 [-52.43, 18.06]	2	8	62	64	13
2	ghazard	A_b20	-7.38 [-52.43, 18.06]	2	8	62	64	0
2	ghazard	B_r50	0.00 [0.00, 0.00]	0	10	0	0	-
2	ghazard	C_b01_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b05_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b10_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	ghazard	C_b20_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	A_b05	-34.17 [-77.09, 3.89]	6	4	124	130	105
2	seismic	A_b10	-34.17 [-77.09, 3.89]	6	4	124	130	79
2	seismic	A_b20	-34.17 [-77.09, 3.89]	6	4	124	130	27
2	seismic	B_r50	0.00 [0.00, 0.00]	0	10	0	0	-
2	seismic	C_b01_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b05_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b10_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismic	C_b20_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	B_r50	0.00 [0.00, 0.00]	0	10	0	0	-
2	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b05_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b10_r50	0.00 [0.00, 0.00]	0	10	0	0	0
2	seismoacoustic	C_b20_r50	0.00 [0.00, 0.00]	0	10	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
3	XGBoost	A_b01	1.55 [0.19, 3.68]	0	24	8	8	3
3	XGBoost	A_b05	-48.64 [-88.57, -19.19]	6	18	49	55	30
3	XGBoost	A_b10	-55.43 [-104.07, -19.18]	7	17	64	71	20
3	XGBoost	A_b20	-65.89 [-120.17, -13.16]	10	14	160	170	67
3	XGBoost	B_r50	-90.31 [-158.34, -28.87]	14	10	234	248	-
3	XGBoost	C_b01_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b05_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b10_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	XGBoost	C_b20_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	A_b20	-13.37 [-35.08, 6.40]	3	21	81	84	0
3	ghazard	B_r50	-137.21 [-236.05, -38.37]	24	0	492	516	-
3	ghazard	C_b01_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b05_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b10_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	ghazard	C_b20_r50	-13.37 [-35.08, 6.40]	3	21	81	84	0
3	seismic	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	A_b20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	B_r50	-66.67 [-137.40, -6.00]	12	12	256	268	-
3	seismic	C_b01_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b05_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b10_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismic	C_b20_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	B_r50	-137.21 [-236.05, -38.37]	24	0	492	516	-

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
3	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b05_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b10_r50	0.00 [0.00, 0.00]	0	24	0	0	0
3	seismoacoustic	C_b20_r50	0.00 [0.00, 0.00]	0	24	0	0	0
4	XGBoost	A_b01	1.16 [0.39, 2.33]	0	18	6	6	1
4	XGBoost	A_b05	-23.06 [-69.19, 5.81]	3	15	31	34	9
4	XGBoost	A_b10	-29.26 [-74.81, 2.52]	4	14	49	53	2
4	XGBoost	A_b20	-47.48 [-108.14, -1.74]	7	11	105	112	9
4	XGBoost	B_r50	-72.09 [-151.17, -15.48]	16	2	428	444	-
4	XGBoost	C_b01_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	XGBoost	C_b05_r50	-6.59 [-27.33, 4.46]	1	17	16	17	0
4	XGBoost	C_b10_r50	-29.26 [-74.81, 2.52]	4	14	49	53	2
4	XGBoost	C_b20_r50	-31.40 [-87.41, 9.50]	5	13	88	93	0
4	ghazard	A_b01	5.43 [0.00, 15.70]	0	18	28	28	23
4	ghazard	A_b05	5.43 [0.00, 15.70]	0	18	28	28	3
4	ghazard	A_b10	5.43 [0.00, 15.70]	0	18	28	28	0
4	ghazard	A_b20	-8.72 [-43.80, 12.40]	2	16	55	57	0
4	ghazard	B_r50	-77.91 [-166.86, -8.72]	18	0	498	516	-
4	ghazard	C_b01_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b05_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b10_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	ghazard	C_b20_r50	-8.72 [-43.80, 12.40]	2	16	55	57	0
4	seismic	A_b01	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	A_b20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	B_r50	-77.91 [-166.86, -8.72]	18	0	498	516	-
4	seismic	C_b01_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b05_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b10_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismic	C_b20_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	18	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
4	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	B_r50	-77.91 [-166.86, -8.72]	18	0	498	516	-
4	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b05_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b10_r50	0.00 [0.00, 0.00]	0	18	0	0	0
4	seismoacoustic	C_b20_r50	0.00 [0.00, 0.00]	0	18	0	0	0
Pooled	XGBoost	A_b01	2.86 [1.75, 3.69]	0	88	59	59	39
Pooled	XGBoost	A_b05	-14.20 [-29.33, -3.49]	9	79	157	166	81
Pooled	XGBoost	A_b10	-22.59 [-43.92, -8.82]	14	74	234	248	84
Pooled	XGBoost	A_b20	-34.80 [-58.17, -12.99]	25	63	532	557	227
Pooled	XGBoost	B_r50	-105.43 [-169.03, -53.50]	64	24	1025	1089	-
Pooled	XGBoost	C_b01_r50	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	XGBoost	C_b05_r50	-1.26 [-6.16, 1.65]	1	87	24	25	0
Pooled	XGBoost	C_b10_r50	-6.79 [-17.70, 0.82]	4	84	60	64	2
Pooled	XGBoost	C_b20_r50	-9.50 [-24.24, 0.73]	6	82	104	110	0
Pooled	ghazard	A_b01	-7.27 [-22.59, 3.25]	4	84	50	54	44
Pooled	ghazard	A_b05	-7.27 [-22.59, 3.25]	4	84	50	54	4
Pooled	ghazard	A_b10	-9.11 [-27.15, 4.17]	6	82	112	118	13
Pooled	ghazard	A_b20	-16.00 [-36.31, -0.14]	11	77	220	231	0
Pooled	ghazard	B_r50	-117.79 [-187.01, -60.93]	78	10	1470	1548	-
Pooled	ghazard	C_b01_r50	-8.63 [-23.36, 1.36]	4	84	22	26	21
Pooled	ghazard	C_b05_r50	-8.63 [-23.36, 1.36]	4	84	22	26	1
Pooled	ghazard	C_b10_r50	-8.63 [-23.36, 1.36]	4	84	22	26	0
Pooled	ghazard	C_b20_r50	-14.15 [-32.62, -0.29]	9	79	158	167	0
Pooled	seismic	A_b01	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	A_b05	-8.53 [-18.42, 0.97]	6	82	124	130	105
Pooled	seismic	A_b10	-8.53 [-18.42, 0.97]	6	82	124	130	79
Pooled	seismic	A_b20	-8.53 [-18.42, 0.97]	6	82	124	130	27
Pooled	seismic	B_r50	-100.15 [-167.09, -45.95]	66	22	1234	1300	-
Pooled	seismic	C_b01_r50	0.00 [0.00, 0.00]	0	88	0	0	0

Block	Method	Rule	Delta loss / 100 [95% CI]	TP	FN	FP	Alerts	Excess
Pooled	seismic	C_b05_r50	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b10_r50	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismic	C_b20_r50	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b01	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b05	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b10	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	A_b20	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	B_r50	-117.79 [-187.01, -60.93]	78	10	1470	1548	-
Pooled	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b05_r50	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b10_r50	0.00 [0.00, 0.00]	0	88	0	0	0
Pooled	seismoacoustic	C_b20_r50	0.00 [0.00, 0.00]	0	88	0	0	0

Fixed XGBoost minus each grade rule

Table S5. XGBoost minus direct grade; assumed missed-hazard cost r=5, false-alert cost 1.

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
1	ghazard	A_b01	1.16 [-4.26, 7.95]
1	ghazard	A_b05	1.55 [-4.07, 8.72]
1	ghazard	A_b10	1.74 [-3.69, 8.91]
1	ghazard	A_b20	1.36 [-3.49, 6.98]
1	ghazard	B_r05	-54.84 [-76.55, -28.87]
1	ghazard	C_b01_r05	-0.39 [-5.62, 6.20]
1	ghazard	C_b05_r05	1.16 [-4.26, 7.95]
1	ghazard	C_b10_r05	1.16 [-4.26, 7.95]
1	ghazard	C_b20_r05	1.74 [-3.68, 8.14]
1	seismic	A_b01	1.55 [0.39, 3.10]
1	seismic	A_b05	1.94 [0.58, 3.88]
1	seismic	A_b10	2.13 [0.78, 4.07]
1	seismic	A_b20	1.74 [-0.58, 4.26]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
1	seismic	B_r05	4.07 [-3.88, 14.34]
1	seismic	C_b01_r05	0.00 [0.00, 0.00]
1	seismic	C_b05_r05	1.55 [0.39, 3.10]
1	seismic	C_b10_r05	1.55 [0.39, 3.10]
1	seismic	C_b20_r05	2.13 [-0.78, 4.85]
1	seismoacoustic	A_b01	1.55 [0.39, 3.10]
1	seismoacoustic	A_b05	1.94 [0.58, 3.88]
1	seismoacoustic	A_b10	2.13 [0.78, 4.07]
1	seismoacoustic	A_b20	1.74 [-0.58, 4.26]
1	seismoacoustic	B_r05	-17.64 [-31.78, -5.43]
1	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]
1	seismoacoustic	C_b05_r05	1.55 [0.39, 3.10]
1	seismoacoustic	C_b10_r05	1.55 [0.39, 3.10]
1	seismoacoustic	C_b20_r05	2.13 [-0.78, 4.85]
2	ghazard	A_b01	7.18 [3.50, 9.51]
2	ghazard	A_b05	13.01 [7.18, 17.28]
2	ghazard	A_b10	8.35 [-4.27, 20.78]
2	ghazard	A_b20	32.23 [19.22, 49.51]
2	ghazard	B_r05	0.00 [0.00, 0.00]
2	ghazard	C_b01_r05	0.00 [0.00, 0.00]
2	ghazard	C_b05_r05	0.00 [0.00, 0.00]
2	ghazard	C_b10_r05	0.00 [0.00, 0.00]
2	ghazard	C_b20_r05	0.00 [0.00, 0.00]
2	seismic	A_b01	7.18 [3.50, 9.51]
2	seismic	A_b05	-5.24 [-15.53, 3.11]
2	seismic	A_b10	0.19 [-9.13, 8.16]
2	seismic	A_b20	24.08 [17.67, 33.98]
2	seismic	B_r05	0.00 [0.00, 0.00]
2	seismic	C_b01_r05	0.00 [0.00, 0.00]
2	seismic	C_b05_r05	0.00 [0.00, 0.00]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
2	seismic	C_b10_r05	0.00 [0.00, 0.00]
2	seismic	C_b20_r05	0.00 [0.00, 0.00]
2	seismoacoustic	A_b01	7.18 [3.50, 9.51]
2	seismoacoustic	A_b05	13.01 [7.18, 17.28]
2	seismoacoustic	A_b10	18.45 [10.29, 25.24]
2	seismoacoustic	A_b20	42.33 [33.20, 53.79]
2	seismoacoustic	B_r05	0.00 [0.00, 0.00]
2	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]
2	seismoacoustic	C_b05_r05	0.00 [0.00, 0.00]
2	seismoacoustic	C_b10_r05	0.00 [0.00, 0.00]
2	seismoacoustic	C_b20_r05	0.00 [0.00, 0.00]
3	ghazard	A_b01	1.55 [0.19, 3.68]
3	ghazard	A_b05	3.68 [-1.36, 6.98]
3	ghazard	A_b10	5.62 [0.39, 8.91]
3	ghazard	A_b20	8.53 [-2.52, 19.19]
3	ghazard	B_r05	0.00 [0.00, 0.00]
3	ghazard	C_b01_r05	0.00 [0.00, 0.00]
3	ghazard	C_b05_r05	0.00 [0.00, 0.00]
3	ghazard	C_b10_r05	0.00 [0.00, 0.00]
3	ghazard	C_b20_r05	0.00 [0.00, 0.00]
3	seismic	A_b01	1.55 [0.19, 3.68]
3	seismic	A_b05	3.68 [-1.36, 6.98]
3	seismic	A_b10	5.62 [0.39, 8.91]
3	seismic	A_b20	21.32 [14.34, 27.33]
3	seismic	B_r05	0.00 [0.00, 0.00]
3	seismic	C_b01_r05	0.00 [0.00, 0.00]
3	seismic	C_b05_r05	0.00 [0.00, 0.00]
3	seismic	C_b10_r05	0.00 [0.00, 0.00]
3	seismic	C_b20_r05	0.00 [0.00, 0.00]
3	seismoacoustic	A_b01	1.55 [0.19, 3.68]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
3	seismoacoustic	A_b05	3.68 [-1.36, 6.98]
3	seismoacoustic	A_b10	5.62 [0.39, 8.91]
3	seismoacoustic	A_b20	21.32 [14.34, 27.33]
3	seismoacoustic	B_r05	0.00 [0.00, 0.00]
3	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]
3	seismoacoustic	C_b05_r05	0.00 [0.00, 0.00]
3	seismoacoustic	C_b10_r05	0.00 [0.00, 0.00]
3	seismoacoustic	C_b20_r05	0.00 [0.00, 0.00]
4	ghazard	A_b01	-4.26 [-14.15, 0.97]
4	ghazard	A_b05	-2.33 [-13.96, 5.23]
4	ghazard	A_b10	0.19 [-10.27, 7.75]
4	ghazard	A_b20	4.84 [-9.11, 14.35]
4	ghazard	B_r05	2.13 [0.00, 4.46]
4	ghazard	C_b01_r05	0.00 [0.00, 0.00]
4	ghazard	C_b05_r05	2.13 [0.00, 4.46]
4	ghazard	C_b10_r05	2.13 [0.00, 4.46]
4	ghazard	C_b20_r05	2.13 [0.00, 4.46]
4	seismic	A_b01	1.16 [0.39, 2.33]
4	seismic	A_b05	3.10 [-0.78, 6.98]
4	seismic	A_b10	5.62 [1.36, 10.86]
4	seismic	A_b20	13.57 [6.01, 20.93]
4	seismic	B_r05	2.13 [0.00, 4.46]
4	seismic	C_b01_r05	0.00 [0.00, 0.00]
4	seismic	C_b05_r05	2.13 [0.00, 4.46]
4	seismic	C_b10_r05	2.13 [0.00, 4.46]
4	seismic	C_b20_r05	2.13 [0.00, 4.46]
4	seismoacoustic	A_b01	1.16 [0.39, 2.33]
4	seismoacoustic	A_b05	3.10 [-0.78, 6.98]
4	seismoacoustic	A_b10	5.62 [1.36, 10.86]
4	seismoacoustic	A_b20	13.57 [6.01, 20.93]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
4	seismoacoustic	B_r05	2.13 [0.00, 4.46]
4	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]
4	seismoacoustic	C_b05_r05	2.13 [0.00, 4.46]
4	seismoacoustic	C_b10_r05	2.13 [0.00, 4.46]
4	seismoacoustic	C_b20_r05	2.13 [0.00, 4.46]
Pooled	ghazard	A_b01	1.41 [-1.60, 3.54]
Pooled	ghazard	A_b05	3.97 [-0.15, 6.50]
Pooled	ghazard	A_b10	3.97 [-0.78, 7.95]
Pooled	ghazard	A_b20	11.73 [6.40, 17.55]
Pooled	ghazard	B_r05	-13.18 [-18.66, -6.69]
Pooled	ghazard	C_b01_r05	-0.10 [-1.41, 1.55]
Pooled	ghazard	C_b05_r05	0.82 [-0.63, 2.81]
Pooled	ghazard	C_b10_r05	0.82 [-0.63, 2.81]
Pooled	ghazard	C_b20_r05	0.97 [-0.48, 2.86]
Pooled	seismic	A_b01	2.86 [1.75, 3.69]
Pooled	seismic	A_b05	0.87 [-2.18, 3.25]
Pooled	seismic	A_b10	3.39 [0.44, 5.77]
Pooled	seismic	A_b20	15.17 [12.21, 18.61]
Pooled	seismic	B_r05	1.55 [-0.63, 4.31]
Pooled	seismic	C_b01_r05	0.00 [0.00, 0.00]
Pooled	seismic	C_b05_r05	0.92 [0.34, 1.65]
Pooled	seismic	C_b10_r05	0.92 [0.34, 1.65]
Pooled	seismic	C_b20_r05	1.07 [0.15, 1.99]
Pooled	seismoacoustic	A_b01	2.86 [1.75, 3.69]
Pooled	seismoacoustic	A_b05	5.43 [2.96, 6.93]
Pooled	seismoacoustic	A_b10	7.95 [4.94, 10.13]
Pooled	seismoacoustic	A_b20	19.73 [16.19, 23.56]
Pooled	seismoacoustic	B_r05	-3.88 [-7.51, -0.77]
Pooled	seismoacoustic	C_b01_r05	0.00 [0.00, 0.00]
Pooled	seismoacoustic	C_b05_r05	0.92 [0.34, 1.65]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
Pooled	seismoacoustic	C_b10_r05	0.92 [0.34, 1.65]
Pooled	seismoacoustic	C_b20_r05	1.07 [0.15, 1.99]

Table S6. XGBoost minus direct grade; assumed missed-hazard cost r=10, false-alert cost 1.

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
1	ghazard	A_b01	5.04 [-4.07, 18.02]
1	ghazard	A_b05	5.43 [-3.88, 18.60]
1	ghazard	A_b10	5.62 [-3.49, 18.99]
1	ghazard	A_b20	3.29 [-3.29, 11.83]
1	ghazard	B_r10	-18.80 [-29.84, -6.01]
1	ghazard	C_b01_r10	3.49 [-5.43, 16.09]
1	ghazard	C_b05_r10	5.04 [-4.07, 18.02]
1	ghazard	C_b10_r10	5.62 [-3.49, 18.99]
1	ghazard	C_b20_r10	4.65 [-3.49, 15.90]
1	seismic	A_b01	1.55 [0.39, 3.10]
1	seismic	A_b05	1.94 [0.58, 3.88]
1	seismic	A_b10	2.13 [0.78, 4.07]
1	seismic	A_b20	-0.19 [-5.23, 3.69]
1	seismic	B_r10	-18.80 [-29.84, -6.01]
1	seismic	C_b01_r10	0.00 [0.00, 0.00]
1	seismic	C_b05_r10	1.55 [0.39, 3.10]
1	seismic	C_b10_r10	2.13 [0.78, 4.07]
1	seismic	C_b20_r10	1.16 [-3.88, 4.84]
1	seismoacoustic	A_b01	1.55 [0.39, 3.10]
1	seismoacoustic	A_b05	1.94 [0.58, 3.88]
1	seismoacoustic	A_b10	2.13 [0.78, 4.07]
1	seismoacoustic	A_b20	-0.19 [-5.23, 3.69]
1	seismoacoustic	B_r10	-18.80 [-29.84, -6.01]
1	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]
1	seismoacoustic	C_b05_r10	1.55 [0.39, 3.10]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
1	seismoacoustic	C_b10_r10	2.13 [0.78, 4.07]
1	seismoacoustic	C_b20_r10	1.16 [-3.88, 4.84]
2	ghazard	A_b01	7.18 [3.50, 9.51]
2	ghazard	A_b05	13.01 [7.18, 17.28]
2	ghazard	A_b10	7.38 [-7.58, 24.27]
2	ghazard	A_b20	28.35 [14.56, 48.75]
2	ghazard	B_r10	0.00 [0.00, 0.00]
2	ghazard	C_b01_r10	0.00 [0.00, 0.00]
2	ghazard	C_b05_r10	0.00 [0.00, 0.00]
2	ghazard	C_b10_r10	0.00 [0.00, 0.00]
2	ghazard	C_b20_r10	0.00 [0.00, 0.00]
2	seismic	A_b01	7.18 [3.50, 9.51]
2	seismic	A_b05	0.58 [-10.49, 8.74]
2	seismic	A_b10	3.11 [-7.58, 10.49]
2	seismic	A_b20	24.08 [16.69, 34.76]
2	seismic	B_r10	0.00 [0.00, 0.00]
2	seismic	C_b01_r10	0.00 [0.00, 0.00]
2	seismic	C_b05_r10	0.00 [0.00, 0.00]
2	seismic	C_b10_r10	0.00 [0.00, 0.00]
2	seismic	C_b20_r10	0.00 [0.00, 0.00]
2	seismoacoustic	A_b01	7.18 [3.50, 9.51]
2	seismoacoustic	A_b05	13.01 [7.18, 17.28]
2	seismoacoustic	A_b10	15.53 [4.47, 23.69]
2	seismoacoustic	A_b20	36.50 [27.18, 49.13]
2	seismoacoustic	B_r10	0.00 [0.00, 0.00]
2	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]
2	seismoacoustic	C_b05_r10	0.00 [0.00, 0.00]
2	seismoacoustic	C_b10_r10	0.00 [0.00, 0.00]
2	seismoacoustic	C_b20_r10	0.00 [0.00, 0.00]
3	ghazard	A_b01	1.55 [0.19, 3.68]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
3	ghazard	A_b05	-2.13 [-10.66, 3.49]
3	ghazard	A_b10	-1.16 [-10.86, 5.04]
3	ghazard	A_b20	1.74 [-11.83, 14.34]
3	ghazard	B_r10	0.00 [0.00, 0.00]
3	ghazard	C_b01_r10	0.00 [0.00, 0.00]
3	ghazard	C_b05_r10	0.00 [0.00, 0.00]
3	ghazard	C_b10_r10	0.00 [0.00, 0.00]
3	ghazard	C_b20_r10	0.00 [0.00, 0.00]
3	seismic	A_b01	1.55 [0.19, 3.68]
3	seismic	A_b05	-2.13 [-10.66, 3.49]
3	seismic	A_b10	-1.16 [-10.86, 5.04]
3	seismic	A_b20	11.63 [0.00, 21.90]
3	seismic	B_r10	0.00 [0.00, 0.00]
3	seismic	C_b01_r10	0.00 [0.00, 0.00]
3	seismic	C_b05_r10	0.00 [0.00, 0.00]
3	seismic	C_b10_r10	0.00 [0.00, 0.00]
3	seismic	C_b20_r10	0.00 [0.00, 0.00]
3	seismoacoustic	A_b01	1.55 [0.19, 3.68]
3	seismoacoustic	A_b05	-2.13 [-10.66, 3.49]
3	seismoacoustic	A_b10	-1.16 [-10.86, 5.04]
3	seismoacoustic	A_b20	11.63 [0.00, 21.90]
3	seismoacoustic	B_r10	0.00 [0.00, 0.00]
3	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]
3	seismoacoustic	C_b05_r10	0.00 [0.00, 0.00]
3	seismoacoustic	C_b10_r10	0.00 [0.00, 0.00]
3	seismoacoustic	C_b20_r10	0.00 [0.00, 0.00]
4	ghazard	A_b01	-4.26 [-14.15, 0.97]
4	ghazard	A_b05	-5.23 [-19.57, 4.65]
4	ghazard	A_b10	-3.68 [-16.87, 6.01]
4	ghazard	A_b20	0.00 [-15.90, 11.82]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
4	ghazard	B_r10	1.16 [-3.10, 4.46]
4	ghazard	C_b01_r10	0.00 [0.00, 0.00]
4	ghazard	C_b05_r10	1.16 [-3.10, 4.46]
4	ghazard	C_b10_r10	1.16 [-3.10, 4.46]
4	ghazard	C_b20_r10	1.16 [-3.10, 4.46]
4	seismic	A_b01	1.16 [0.39, 2.33]
4	seismic	A_b05	0.19 [-7.76, 6.40]
4	seismic	A_b10	1.74 [-6.40, 9.30]
4	seismic	A_b20	6.78 [-5.62, 16.67]
4	seismic	B_r10	1.16 [-3.10, 4.46]
4	seismic	C_b01_r10	0.00 [0.00, 0.00]
4	seismic	C_b05_r10	1.16 [-3.10, 4.46]
4	seismic	C_b10_r10	1.16 [-3.10, 4.46]
4	seismic	C_b20_r10	1.16 [-3.10, 4.46]
4	seismoacoustic	A_b01	1.16 [0.39, 2.33]
4	seismoacoustic	A_b05	0.19 [-7.76, 6.40]
4	seismoacoustic	A_b10	1.74 [-6.40, 9.30]
4	seismoacoustic	A_b20	6.78 [-5.62, 16.67]
4	seismoacoustic	B_r10	1.16 [-3.10, 4.46]
4	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]
4	seismoacoustic	C_b05_r10	1.16 [-3.10, 4.46]
4	seismoacoustic	C_b10_r10	1.16 [-3.10, 4.46]
4	seismoacoustic	C_b20_r10	1.16 [-3.10, 4.46]
Pooled	ghazard	A_b01	2.38 [-1.26, 5.67]
Pooled	ghazard	A_b05	2.76 [-2.76, 6.83]
Pooled	ghazard	A_b10	2.04 [-4.32, 7.61]
Pooled	ghazard	A_b20	8.34 [2.13, 15.32]
Pooled	ghazard	B_r10	-4.41 [-7.03, -0.97]
Pooled	ghazard	C_b01_r10	0.87 [-1.36, 3.97]
Pooled	ghazard	C_b05_r10	1.55 [-0.92, 5.04]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
Pooled	ghazard	C_b10_r10	1.70 [-0.82, 5.19]
Pooled	ghazard	C_b20_r10	1.45 [-0.83, 4.65]
Pooled	seismic	A_b01	2.86 [1.75, 3.69]
Pooled	seismic	A_b05	0.15 [-4.12, 3.01]
Pooled	seismic	A_b10	1.45 [-2.76, 4.27]
Pooled	seismic	A_b20	10.57 [6.20, 15.22]
Pooled	seismic	B_r10	-4.41 [-7.03, -0.97]
Pooled	seismic	C_b01_r10	0.00 [0.00, 0.00]
Pooled	seismic	C_b05_r10	0.68 [-0.34, 1.65]
Pooled	seismic	C_b10_r10	0.82 [-0.19, 1.84]
Pooled	seismic	C_b20_r10	0.58 [-1.02, 1.89]
Pooled	seismoacoustic	A_b01	2.86 [1.75, 3.69]
Pooled	seismoacoustic	A_b05	3.25 [-0.29, 5.38]
Pooled	seismoacoustic	A_b10	4.56 [-0.00, 7.51]
Pooled	seismoacoustic	A_b20	13.67 [8.77, 18.61]
Pooled	seismoacoustic	B_r10	-4.41 [-7.03, -0.97]
Pooled	seismoacoustic	C_b01_r10	0.00 [0.00, 0.00]
Pooled	seismoacoustic	C_b05_r10	0.68 [-0.34, 1.65]
Pooled	seismoacoustic	C_b10_r10	0.82 [-0.19, 1.84]
Pooled	seismoacoustic	C_b20_r10	0.58 [-1.02, 1.89]

Table S7. XGBoost minus direct grade; assumed missed-hazard cost r=20, false-alert cost 1.

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
1	ghazard	A_b01	12.79 [-4.07, 37.40]
1	ghazard	A_b05	13.18 [-3.49, 37.98]
1	ghazard	A_b10	13.37 [-3.49, 38.37]
1	ghazard	A_b20	7.17 [-3.29, 22.49]
1	ghazard	B_r20	-14.92 [-27.71, 1.55]
1	ghazard	C_b01_r20	11.24 [-5.43, 35.47]
1	ghazard	C_b05_r20	12.79 [-4.07, 37.40]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
1	ghazard	C_b10_r20	13.37 [-3.49, 38.37]
1	ghazard	C_b20_r20	10.47 [-3.49, 31.40]
1	seismic	A_b01	1.55 [0.39, 3.10]
1	seismic	A_b05	1.94 [0.58, 3.88]
1	seismic	A_b10	2.13 [0.78, 4.07]
1	seismic	A_b20	-4.07 [-15.12, 3.68]
1	seismic	B_r20	-14.92 [-27.71, 1.55]
1	seismic	C_b01_r20	0.00 [0.00, 0.00]
1	seismic	C_b05_r20	1.55 [0.39, 3.10]
1	seismic	C_b10_r20	2.13 [0.78, 4.07]
1	seismic	C_b20_r20	-0.78 [-10.08, 4.84]
1	seismoacoustic	A_b01	1.55 [0.39, 3.10]
1	seismoacoustic	A_b05	1.94 [0.58, 3.88]
1	seismoacoustic	A_b10	2.13 [0.78, 4.07]
1	seismoacoustic	A_b20	-4.07 [-15.12, 3.68]
1	seismoacoustic	B_r20	-14.92 [-27.71, 1.55]
1	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]
1	seismoacoustic	C_b05_r20	1.55 [0.39, 3.10]
1	seismoacoustic	C_b10_r20	2.13 [0.78, 4.07]
1	seismoacoustic	C_b20_r20	-0.78 [-10.08, 4.84]
2	ghazard	A_b01	7.18 [3.50, 9.51]
2	ghazard	A_b05	13.01 [7.18, 17.28]
2	ghazard	A_b10	5.44 [-18.07, 32.24]
2	ghazard	A_b20	20.58 [0.97, 49.32]
2	ghazard	B_r20	0.00 [0.00, 0.00]
2	ghazard	C_b01_r20	0.00 [0.00, 0.00]
2	ghazard	C_b05_r20	0.00 [0.00, 0.00]
2	ghazard	C_b10_r20	0.00 [0.00, 0.00]
2	ghazard	C_b20_r20	0.00 [0.00, 0.00]
2	seismic	A_b01	7.18 [3.50, 9.51]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
2	seismic	A_b05	12.23 [-3.88, 27.38]
2	seismic	A_b10	8.93 [-7.97, 19.03]
2	seismic	A_b20	24.08 [12.23, 37.67]
2	seismic	B_r20	0.00 [0.00, 0.00]
2	seismic	C_b01_r20	0.00 [0.00, 0.00]
2	seismic	C_b05_r20	0.00 [0.00, 0.00]
2	seismic	C_b10_r20	0.00 [0.00, 0.00]
2	seismic	C_b20_r20	0.00 [0.00, 0.00]
2	seismoacoustic	A_b01	7.18 [3.50, 9.51]
2	seismoacoustic	A_b05	13.01 [7.18, 17.28]
2	seismoacoustic	A_b10	9.71 [-9.32, 22.52]
2	seismoacoustic	A_b20	24.85 [9.13, 43.50]
2	seismoacoustic	B_r20	0.00 [0.00, 0.00]
2	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]
2	seismoacoustic	C_b05_r20	0.00 [0.00, 0.00]
2	seismoacoustic	C_b10_r20	0.00 [0.00, 0.00]
2	seismoacoustic	C_b20_r20	0.00 [0.00, 0.00]
3	ghazard	A_b01	1.55 [0.19, 3.68]
3	ghazard	A_b05	-13.76 [-30.43, -2.13]
3	ghazard	A_b10	-14.73 [-34.30, -1.35]
3	ghazard	A_b20	-11.82 [-33.53, 8.91]
3	ghazard	B_r20	0.00 [0.00, 0.00]
3	ghazard	C_b01_r20	0.00 [0.00, 0.00]
3	ghazard	C_b05_r20	0.00 [0.00, 0.00]
3	ghazard	C_b10_r20	0.00 [0.00, 0.00]
3	ghazard	C_b20_r20	0.00 [0.00, 0.00]
3	seismic	A_b01	1.55 [0.19, 3.68]
3	seismic	A_b05	-13.76 [-30.43, -2.13]
3	seismic	A_b10	-14.73 [-34.30, -1.35]
3	seismic	A_b20	-7.75 [-29.46, 12.21]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
3	seismic	B_r20	-3.10 [-26.94, 24.23]
3	seismic	C_b01_r20	0.00 [0.00, 0.00]
3	seismic	C_b05_r20	0.00 [0.00, 0.00]
3	seismic	C_b10_r20	0.00 [0.00, 0.00]
3	seismic	C_b20_r20	0.00 [0.00, 0.00]
3	seismoacoustic	A_b01	1.55 [0.19, 3.68]
3	seismoacoustic	A_b05	-13.76 [-30.43, -2.13]
3	seismoacoustic	A_b10	-14.73 [-34.30, -1.35]
3	seismoacoustic	A_b20	-7.75 [-29.46, 12.21]
3	seismoacoustic	B_r20	0.00 [0.00, 0.00]
3	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]
3	seismoacoustic	C_b05_r20	0.00 [0.00, 0.00]
3	seismoacoustic	C_b10_r20	0.00 [0.00, 0.00]
3	seismoacoustic	C_b20_r20	0.00 [0.00, 0.00]
4	ghazard	A_b01	-4.26 [-14.15, 0.97]
4	ghazard	A_b05	-11.05 [-32.75, 3.88]
4	ghazard	A_b10	-11.43 [-32.56, 4.07]
4	ghazard	A_b20	-9.69 [-33.53, 7.37]
4	ghazard	B_r20	-13.57 [-32.76, 12.80]
4	ghazard	C_b01_r20	0.00 [0.00, 0.00]
4	ghazard	C_b05_r20	-0.78 [-9.69, 4.46]
4	ghazard	C_b10_r20	-6.01 [-22.68, 7.37]
4	ghazard	C_b20_r20	-6.01 [-22.68, 7.37]
4	seismic	A_b01	1.16 [0.39, 2.33]
4	seismic	A_b05	-5.62 [-22.67, 5.81]
4	seismic	A_b10	-6.01 [-22.68, 7.37]
4	seismic	A_b20	-6.78 [-30.43, 11.63]
4	seismic	B_r20	-13.57 [-32.76, 12.80]
4	seismic	C_b01_r20	0.00 [0.00, 0.00]
4	seismic	C_b05_r20	-0.78 [-9.69, 4.46]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
4	seismic	C_b10_r20	-6.01 [-22.68, 7.37]
4	seismic	C_b20_r20	-6.01 [-22.68, 7.37]
4	seismoacoustic	A_b01	1.16 [0.39, 2.33]
4	seismoacoustic	A_b05	-5.62 [-22.67, 5.81]
4	seismoacoustic	A_b10	-6.01 [-22.68, 7.37]
4	seismoacoustic	A_b20	-6.78 [-30.43, 11.63]
4	seismoacoustic	B_r20	10.66 [-15.51, 31.98]
4	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]
4	seismoacoustic	C_b05_r20	-0.78 [-9.69, 4.46]
4	seismoacoustic	C_b10_r20	-6.01 [-22.68, 7.37]
4	seismoacoustic	C_b20_r20	-6.01 [-22.68, 7.37]
Pooled	ghazard	A_b01	4.31 [-0.92, 10.52]
Pooled	ghazard	A_b05	0.34 [-8.58, 8.05]
Pooled	ghazard	A_b10	-1.84 [-12.80, 7.71]
Pooled	ghazard	A_b20	1.55 [-8.24, 11.98]
Pooled	ghazard	B_r20	-7.13 [-12.31, 1.07]
Pooled	ghazard	C_b01_r20	2.81 [-1.36, 8.82]
Pooled	ghazard	C_b05_r20	3.01 [-1.70, 9.65]
Pooled	ghazard	C_b10_r20	1.84 [-4.51, 8.97]
Pooled	ghazard	C_b20_r20	1.11 [-4.75, 7.37]
Pooled	seismic	A_b01	2.86 [1.75, 3.69]
Pooled	seismic	A_b05	-1.31 [-8.97, 3.98]
Pooled	seismic	A_b10	-2.42 [-10.66, 2.33]
Pooled	seismic	A_b20	1.36 [-7.03, 9.21]
Pooled	seismic	B_r20	-7.90 [-15.90, 2.76]
Pooled	seismic	C_b01_r20	0.00 [0.00, 0.00]
Pooled	seismic	C_b05_r20	0.19 [-1.79, 1.65]
Pooled	seismic	C_b10_r20	-0.97 [-5.14, 2.18]
Pooled	seismic	C_b20_r20	-1.70 [-6.69, 1.84]
Pooled	seismoacoustic	A_b01	2.86 [1.75, 3.69]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
Pooled	seismoacoustic	A_b05	-1.11 [-7.46, 3.06]
Pooled	seismoacoustic	A_b10	-2.23 [-10.95, 3.30]
Pooled	seismoacoustic	A_b20	1.55 [-7.51, 10.23]
Pooled	seismoacoustic	B_r20	-1.07 [-8.24, 5.72]
Pooled	seismoacoustic	C_b01_r20	0.00 [0.00, 0.00]
Pooled	seismoacoustic	C_b05_r20	0.19 [-1.79, 1.65]
Pooled	seismoacoustic	C_b10_r20	-0.97 [-5.14, 2.18]
Pooled	seismoacoustic	C_b20_r20	-1.70 [-6.69, 1.84]

Table S8. XGBoost minus direct grade; assumed missed-hazard cost r=50, false-alert cost 1.

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
1	ghazard	A_b01	36.05 [-4.07, 95.54]
1	ghazard	A_b05	36.43 [-3.49, 96.12]
1	ghazard	A_b10	36.63 [-3.49, 96.51]
1	ghazard	A_b20	18.80 [-3.49, 57.17]
1	ghazard	B_r50	-3.29 [-24.42, 28.88]
1	ghazard	C_b01_r50	34.50 [-5.43, 93.60]
1	ghazard	C_b05_r50	36.05 [-4.07, 95.54]
1	ghazard	C_b10_r50	36.63 [-3.49, 96.51]
1	ghazard	C_b20_r50	27.91 [-3.49, 77.91]
1	seismic	A_b01	1.55 [0.39, 3.10]
1	seismic	A_b05	1.94 [0.58, 3.88]
1	seismic	A_b10	2.13 [0.78, 4.07]
1	seismic	A_b20	-15.70 [-44.19, 3.68]
1	seismic	B_r50	-3.29 [-24.42, 28.88]
1	seismic	C_b01_r50	0.00 [0.00, 0.00]
1	seismic	C_b05_r50	1.55 [0.39, 3.10]
1	seismic	C_b10_r50	2.13 [0.78, 4.07]
1	seismic	C_b20_r50	-6.59 [-27.52, 4.84]
1	seismoacoustic	A_b01	1.55 [0.39, 3.10]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
1	seismoacoustic	A_b05	1.94 [0.58, 3.88]
1	seismoacoustic	A_b10	2.13 [0.78, 4.07]
1	seismoacoustic	A_b20	-15.70 [-44.19, 3.68]
1	seismoacoustic	B_r50	-3.29 [-24.42, 28.88]
1	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]
1	seismoacoustic	C_b05_r50	1.55 [0.39, 3.10]
1	seismoacoustic	C_b10_r50	2.13 [0.78, 4.07]
1	seismoacoustic	C_b20_r50	-6.59 [-27.52, 4.84]
2	ghazard	A_b01	7.18 [3.50, 9.51]
2	ghazard	A_b05	13.01 [7.18, 17.28]
2	ghazard	A_b10	-0.39 [-54.56, 60.40]
2	ghazard	A_b20	-2.72 [-50.29, 58.06]
2	ghazard	B_r50	0.00 [0.00, 0.00]
2	ghazard	C_b01_r50	0.00 [0.00, 0.00]
2	ghazard	C_b05_r50	0.00 [0.00, 0.00]
2	ghazard	C_b10_r50	0.00 [0.00, 0.00]
2	ghazard	C_b20_r50	0.00 [0.00, 0.00]
2	seismic	A_b01	7.18 [3.50, 9.51]
2	seismic	A_b05	47.18 [4.66, 91.86]
2	seismic	A_b10	26.41 [-15.34, 54.57]
2	seismic	A_b20	24.08 [-4.28, 52.62]
2	seismic	B_r50	0.00 [0.00, 0.00]
2	seismic	C_b01_r50	0.00 [0.00, 0.00]
2	seismic	C_b05_r50	0.00 [0.00, 0.00]
2	seismic	C_b10_r50	0.00 [0.00, 0.00]
2	seismic	C_b20_r50	0.00 [0.00, 0.00]
2	seismoacoustic	A_b01	7.18 [3.50, 9.51]
2	seismoacoustic	A_b05	13.01 [7.18, 17.28]
2	seismoacoustic	A_b10	-7.77 [-53.40, 21.75]
2	seismoacoustic	A_b20	-10.10 [-53.79, 34.56]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
2	seismoacoustic	B_r50	0.00 [0.00, 0.00]
2	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]
2	seismoacoustic	C_b05_r50	0.00 [0.00, 0.00]
2	seismoacoustic	C_b10_r50	0.00 [0.00, 0.00]
2	seismoacoustic	C_b20_r50	0.00 [0.00, 0.00]
3	ghazard	A_b01	1.55 [0.19, 3.68]
3	ghazard	A_b05	-48.64 [-88.57, -19.19]
3	ghazard	A_b10	-55.43 [-104.07, -19.18]
3	ghazard	A_b20	-52.52 [-106.21, -3.29]
3	ghazard	B_r50	46.90 [-6.21, 98.85]
3	ghazard	C_b01_r50	0.00 [0.00, 0.00]
3	ghazard	C_b05_r50	0.00 [0.00, 0.00]
3	ghazard	C_b10_r50	0.00 [0.00, 0.00]
3	ghazard	C_b20_r50	13.37 [-6.40, 35.08]
3	seismic	A_b01	1.55 [0.19, 3.68]
3	seismic	A_b05	-48.64 [-88.57, -19.19]
3	seismic	A_b10	-55.43 [-104.07, -19.18]
3	seismic	A_b20	-65.89 [-120.17, -13.16]
3	seismic	B_r50	-23.64 [-78.88, 18.22]
3	seismic	C_b01_r50	0.00 [0.00, 0.00]
3	seismic	C_b05_r50	0.00 [0.00, 0.00]
3	seismic	C_b10_r50	0.00 [0.00, 0.00]
3	seismic	C_b20_r50	0.00 [0.00, 0.00]
3	seismoacoustic	A_b01	1.55 [0.19, 3.68]
3	seismoacoustic	A_b05	-48.64 [-88.57, -19.19]
3	seismoacoustic	A_b10	-55.43 [-104.07, -19.18]
3	seismoacoustic	A_b20	-65.89 [-120.17, -13.16]
3	seismoacoustic	B_r50	46.90 [-6.21, 98.85]
3	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]
3	seismoacoustic	C_b05_r50	0.00 [0.00, 0.00]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
3	seismoacoustic	C_b10_r50	0.00 [0.00, 0.00]
3	seismoacoustic	C_b20_r50	0.00 [0.00, 0.00]
4	ghazard	A_b01	-4.26 [-14.15, 0.97]
4	ghazard	A_b05	-28.49 [-76.76, 3.88]
4	ghazard	A_b10	-34.69 [-82.95, -0.78]
4	ghazard	A_b20	-38.76 [-93.61, -2.13]
4	ghazard	B_r50	5.81 [-14.73, 33.53]
4	ghazard	C_b01_r50	0.00 [0.00, 0.00]
4	ghazard	C_b05_r50	-6.59 [-27.33, 4.46]
4	ghazard	C_b10_r50	-29.26 [-74.81, 2.52]
4	ghazard	C_b20_r50	-22.67 [-67.66, 7.95]
4	seismic	A_b01	1.16 [0.39, 2.33]
4	seismic	A_b05	-23.06 [-69.19, 5.81]
4	seismic	A_b10	-29.26 [-74.81, 2.52]
4	seismic	A_b20	-47.48 [-108.14, -1.74]
4	seismic	B_r50	5.81 [-14.73, 33.53]
4	seismic	C_b01_r50	0.00 [0.00, 0.00]
4	seismic	C_b05_r50	-6.59 [-27.33, 4.46]
4	seismic	C_b10_r50	-29.26 [-74.81, 2.52]
4	seismic	C_b20_r50	-31.40 [-87.41, 9.50]
4	seismoacoustic	A_b01	1.16 [0.39, 2.33]
4	seismoacoustic	A_b05	-23.06 [-69.19, 5.81]
4	seismoacoustic	A_b10	-29.26 [-74.81, 2.52]
4	seismoacoustic	A_b20	-47.48 [-108.14, -1.74]
4	seismoacoustic	B_r50	5.81 [-14.73, 33.53]
4	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]
4	seismoacoustic	C_b05_r50	-6.59 [-27.33, 4.46]
4	seismoacoustic	C_b10_r50	-29.26 [-74.81, 2.52]
4	seismoacoustic	C_b20_r50	-31.40 [-87.41, 9.50]
Pooled	ghazard	A_b01	10.13 [-0.87, 25.11]

Block	Grade field	Rule	ML minus grade loss / 100 [95% CI]
Pooled	ghazard	A_b05	-6.93 [-26.32, 12.41]
Pooled	ghazard	A_b10	-13.48 [-38.69, 9.60]
Pooled	ghazard	A_b20	-18.81 [-41.40, 5.19]
Pooled	ghazard	B_r50	12.36 [-2.52, 30.30]
Pooled	ghazard	C_b01_r50	8.63 [-1.36, 23.36]
Pooled	ghazard	C_b05_r50	7.37 [-4.41, 24.24]
Pooled	ghazard	C_b10_r50	1.84 [-13.58, 19.25]
Pooled	ghazard	C_b20_r50	4.65 [-10.76, 20.89]
Pooled	seismic	A_b01	2.86 [1.75, 3.69]
Pooled	seismic	A_b05	-5.67 [-25.69, 8.00]
Pooled	seismic	A_b10	-14.06 [-35.15, -1.94]
Pooled	seismic	A_b20	-26.27 [-48.23, -7.85]
Pooled	seismic	B_r50	-5.28 [-20.36, 10.08]
Pooled	seismic	C_b01_r50	0.00 [0.00, 0.00]
Pooled	seismic	C_b05_r50	-1.26 [-6.16, 1.65]
Pooled	seismic	C_b10_r50	-6.79 [-17.70, 0.82]
Pooled	seismic	C_b20_r50	-9.50 [-24.24, 0.73]
Pooled	seismoacoustic	A_b01	2.86 [1.75, 3.69]
Pooled	seismoacoustic	A_b05	-14.20 [-29.33, -3.49]
Pooled	seismoacoustic	A_b10	-22.59 [-43.92, -8.82]
Pooled	seismoacoustic	A_b20	-34.80 [-58.17, -12.99]
Pooled	seismoacoustic	B_r50	12.36 [-2.52, 30.30]
Pooled	seismoacoustic	C_b01_r50	0.00 [0.00, 0.00]
Pooled	seismoacoustic	C_b05_r50	-1.26 [-6.16, 1.65]
Pooled	seismoacoustic	C_b10_r50	-6.79 [-17.70, 0.82]
Pooled	seismoacoustic	C_b20_r50	-9.50 [-24.24, 0.73]